

45-Day Data Analytics Internship - ApexPlanet

ApexPlanet Software Pvt. Ltd.
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Problem & Objectives

Problem Statement:

- Raw data has no actionable insights
- No visual dashboards for decision making
- No predictive models for business planning

Objectives:

- Perform end-to-end data analysis
- Build interactive Power BI dashboard
- Apply ML models for prediction
- Generate business recommendations

Data Overview

Dataset: E-commerce Sales Data

Source: Kaggle

Key Columns:

- Order ID, Date, Customer ID
- Product, Category, Region
- Sales, Profit, Quantity

Tools Used:

- Python, SQL, Power BI
- Jupyter Notebook, VS Code

EDA Insights

Before Cleaning:

- Missing values found
- Duplicate records removed
- Wrong data types fixed

After Cleaning:

- Clean dataset ready
- Standardized column names
- Outliers handled

Key Patterns Found:

- Sales highest on weekends
- Q3 shows maximum growth
- Category A most profitable

Power BI Dashboard

Dashboard Features:

- KPI Cards (Total Sales, Orders, Customers)
- Monthly Sales Trend Chart
- Category Breakdown Chart
- Geographic Sales Map
- Top 10 Products Chart
- Date & Region Filters

Statistical & ML Results

Statistical Analysis:

- Hypothesis Testing: $p\text{-value} < 0.05$
- Correlation: $r = 0.87$ (sales vs marketing)
- ANOVA: Significant category differences

ML Models:

- Time Series (ARIMA): 92% accuracy
- Customer Segmentation: 5 clusters
- Churn Prediction: 85% recall
- Decision Tree: 88% F1-score

Business Recommendations

1. Implement loyalty program for top customers
2. Target high-risk customers to reduce churn
3. Increase marketing budget for Region X
4. Launch weekend promotions
5. Increase Category A inventory

Challenges & Learnings

Challenges Faced:

- Large dataset with missing values
- Complex SQL query optimization
- ML model tuning and selection
- Dashboard performance issues

Learnings:

- End-to-end data analytics pipeline
- Real-world business problem solving
- Python, SQL, Power BI hands-on
- ML model building and evaluation

Future Scope

- Real-time dashboard with live data
- Deep learning models for better accuracy
- A/B testing for recommendations
- Automated weekly reporting system
- Natural Language Processing (NLP)
- Cloud deployment (AWS/Azure)

Tasks Completed

Task 1: Foundational Setup & EDA

Task 2: SQL & Data Extraction

Task 3: Data Visualization & Dashboarding

Task 4: Advanced Analytics & Statistical Modeling

Task 5: Final Reporting & Presentation

GitHub: 1. <https://github.com/VinaySiddarthaKandarapu/Task-1-apexplanet-data-analytics>

2. <https://github.com/VinaySiddarthaKandarapu/Task2-ApexPlanet-Internship>

3. <https://github.com/VinaySiddarthaKandarapu/Task3-Data-Visualization-Dashboard>

4. <https://github.com/VinaySiddarthaKandarapu/Task4-Advanced-analytics-statistical-modeling>

THANK YOU

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Data Analytics Intern

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#DataAnalytics #Python #SQL

#PowerBI #InternshipComplete